



Different aspects of *Pseudomonas aeruginosa* biofilm: an in-depth analysis from formation to detection

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Abstract

Planktonic pathogens pose a significant threat to the global community, especially as they develop into microbial biofilms. *Pseudomonas aeruginosa*, a gram-negative bacillus, is notorious for causing nosocomial infections and can easily transform into a multidrug-resistant form, particularly within biofilms. This transformation leads to chronic infections in immunocompromised individuals, increasing morbidity and mortality rates worldwide. *P. aeruginosa* uses biofilms to survive, becoming robust and evading the host's immune protection system. Biofilms also shield microorganisms from antibiotics, rendering them drug-resistant and causing treatment failures. *P. aeruginosa* biofilms are often associated with device-related infections, as indwelling medical devices (IMDs) serve as breeding grounds for microbial biofilm growth. The ubiquitous presence of *P. aeruginosa* on both living and abiotic surfaces makes it an excellent model for studying biofilms. The virulence factors of *P. aeruginosa*, such as lipopolysaccharides, pili, flagella, adhesins, and exopolysaccharides, facilitate disease progression and contribute to biofilm development and maturation. This review focuses on analyzing the characteristics of *P. aeruginosa* biofilms, including antimicrobial resistance and gene regulation. Methods for detecting biofilms are also briefly described. Additionally, this review provides a comprehensive exploration of the most effective methods for preventing and controlling microbial biofilms, particularly those formed by *P. aeruginosa*. This study aims to inspire future research and the development of innovative technologies to resist microbial adherence, colonization, and biofilm formation.

Keywords Biofilm · *Pseudomonas aeruginosa* · Quorum sensing · Antibiotic susceptibility

Introduction

Over the past two decades, microbial biofilm has been intensively studied to understand the behavior of bacteria in multiple communities and the ongoing infections associated with

biofilm in industrial and medical fields (Jones et al. 1969; Characklis 1973; Costerton et al. 1978). Planktonic bacteria in biofilm communities are more supreme in causing chronic infections than their free-living counterparts because bacteria in such communities can easily acquire resistance against antibiotics and biocide treatment and escape the host immune response (Hänsch et al. 2012; Sharma et al. 2023;

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Liu et al. 2024). It was the first AV Leeuwenhoek to observe the biofilm as particles “scraped off his tongue” in “scurf of the teeth” with his primitive microscope (Dobell 1960). After the observation of biofilm by Louis Pasteur as sketched aggregates of bacteria, it remained mundane and unknown for a century (Hoiby 2014). In 1985, Costerton demonstrated the increased antimicrobial resistance in biofilm compared to planktonic bacteria in medical devices (Nickel et al. 1985). A biofilm is a conglomerate aggregation of microbial species irreversibly bound to biotic or abiotic surfaces, secreting exopolysaccharides (Flemming et al. 2010). The outermost surface film, the biofilm base, and the innermost conditioning layer form the biofilm altogether (Ghazay et al. 2021).

- a. Surface film: in this outermost layer, the free-floating microbes are released and spread through the different compartments.
- b. Biofilm base: comprising microorganisms, this layer is present amidst the surface film and conditioning layer.
- c. Conditioning/linking layer: this layer is affixed to the biotic or abiotic surface, such as tissues and biomaterials.

The biofilm formation is mainly due to signaling pathways initiated by cell-to-cell communications. This phenomenon is known as quorum sensing (QS) (Sahreen et al. 2022). Furthermore, quorum sensing also participates in the maturation of biofilm by controlling it, which results in persistent or chronic infection (Preda et al. 2019). Jovchevski et al. (2022) reported that during an altered environment in terms of temperature, oxygen level, and pH, bacteria tend to communicate cell-to-cell through chemical signaling quorum sensing molecules in biofilm communities. In these communities, microbes are well protected from the adverse effects of antibiotics as the biofilm encourages the perseverance of microorganisms and shields them from any unfavorable factors such as penetration of antibiotics (Uruén et al. 2020; Liu et al. 2024). The microbial subpopulations within a biofilm are called persister variants, typically dormant cells with deficient metabolic activity (Niu et al. 2024). Grassi et al. (2017) also suggested that the persister variants lack metabolic activity and replicate inactively, which is why they are less susceptible to antibiotics. Hence, it explains that the significant reason that promotes the colonization of an infection and its recurrence involves biofilm, persister variants, and quorum sensing altogether (Grassi et al. 2017; Niu et al. 2024).

Biofilm on medical devices is 60–70% of nosocomial infections caused either by single bacterial species or a concoction of species (Bryers 2008). It is important to understand biofilm-associated bacteria and the efficiency of these sessile as well as planktonic species to colonize a biofilm. *Pseudomonas aeruginosa*, a metabolically versatile

organism commonly involved in healthcare-acquired infection (Yang et al. 2011) and hugely associated with biofilm, is extensively studied due to its involvement in microbial biofilm and pathogenesis of the biofilm infection. Belonging to the Pseudomonadaceae family, *P. aeruginosa* is not only associated with the environment but also causes various clinical infections like respiratory tract infections, implant infections, prosthetic infections, and burn or wound infection (Mohammadi et al. 2020).

Yetkin et al. (2006) suggested that *P. infection* in hospitals primarily affects patients with chronic illnesses, such as in intensive care units (ICUs), indwelling medical devices (IMDs), or patients in burn wards. *P. aeruginosa* has the potential to withstand a nutritional deficient environment as it requires less nutrition for its survival (Yetkin et al. 2006). It can endure in extreme physical or environmental conditions. These factors, in total, make *P. aeruginosa* infections more persistent and prolonged in communities and hospitals (Lister et al. 2009). The infection caused by *P. aeruginosa* being nosocomial can be explained by invading an immunocompromised host's tissues and seldom infecting healthy tissues (Mesaros et al. 2007).

Cullen et al. (2015) discussed that the colonization, survival, and proliferation of *P. aeruginosa* infection occurring in various environments are primarily due to pathogenicity-adaptive characteristics and virulence mechanisms that form its varied nature. Hence, this opportunistic pathogen has the potential to effortlessly acquire possession of medical devices and tissues and build resistant biofilm, which has evolved into an enormous concern for global health (Del Mar Cendra et al. 2021). Diseases involving cystic fibrosis, cancer, catheters, burns, and wound injuries are a principal threat because of the compromised immunity of patients. AIDS and non-healing diabetic patients with non-healing wounds are also prone to such infections. *P. aeruginosa* was categorized among the most life-threatening bacteria by the World Health Organization in 2017. Therefore, new antibiotics against the infection should be researched and developed on a priority basis (Thi et al. 2020). The prevalence rate of *P. aeruginosa* between 7.1% and 7.3% among nosocomial infections has been discussed by Magill et al. (2014). A variation from 10 to 30% in the prevalence rate of the infection has also been noticed in India (Senthamarai et al. 2014). Bacteria feel more protected in biofilm communities than planktonic ones as they are shielded from host immune responses, antibiotics, and adverse environmental conditions (Sharma et al. 2019).

Fux et al. (2005) added that microbial biofilm can cause significant problems in both medical settings in the form of persistent and recurrent infections or device-related infections. It poses a severe threat to non-medical or industrial settings (Fux et al. 2005). Furthermore, biofilm infection caused by pathogenic strains on indwelling medical devices

is the primary concern to the public. Hence, this has an aversive effect on public health and decreases the drug's efficacy (Lindsay et al. 2006). Bauer et al. (2002) listed biofilm-associated diseases as urinary tract infections associated with catheters (CAUTI), middle ear infections, dental plaque, prosthetic valve endocarditis, and cystic fibrosis. According to Donlan (2001), the development of extraluminal biofilm occurs after 1 week of catheterization, as observed in central venous catheters, which increases the rate of catheter-associated bloodstream infections.

When bacteria reside in microbial biofilm, there is an elevation in its mutation; efflux pumps are upregulated, and metabolic functions are lower, making it a thousand times more resistant to antibiotics than the planktonic microbial cells (Tarawneh et al. 2021). Hence, there is a requisite for the perceptivity and knowledge of the virulence genes and resistance against antibiotics for the apprehension of *P. infections* (Seyed et al. 2021).

Characteristics of *P. aeruginosa*

Ubiquitously present *P. aeruginosa* with a monotrichous flagellum appear singly, in pairs or short chains, producing pink rod-shaped bacilli when seen under gram stain (Azam et al. 2018). “Pseudo” means false, and “monas” means single in Greek, while “aeruginosa” derives from the Latin term ‘aerūgō’, meaning rusted cooper or greenish blue. It was described for the first time in 1882 by French pharmacist Carle Gessard. He referred to it as “On the blue and green coloration of bandages” (Diggle et al. 2020). Later, these green-blue bandages became known as pyocyanin pigment generated by *P. aeruginosa*. Pyocyanin pigment derived from phenazone that actively participates in iron metabolism. It is lipid soluble and maintains the redox equilibrium surrounding the bacteria, providing cell–cell communication (Hall et al. 2016). Furthermore, *P. aeruginosa* may grow anaerobically if nitrates are available. Therefore, they are facultative anaerobes and yield acid from sugars such as glucose, fructose, and xylose, except for lactose or sucrose.

P. aeruginosa can easily survive in natural and artificial environments, including lakes, hospitals, and household sink drains, with the help of its binding appendages, including pili and flagella (Remold et al. 2011). It is not found as a normal commensal of the human microbiota, and its colonization in healthy individuals is rare, especially at moist sites, such as the perineum, axilla, and ear. However, it has been found in the throat, nasal mucosa, or skin of 5–10% of healthy individuals and an elevated prevalence of 2% has been observed in stools (Berthelot et al. 2001). This might be due to the rising resistance and biofilm-forming capability of *P. aeruginosa*. Nosocomial infections caused by this pathogen have emerged as a global

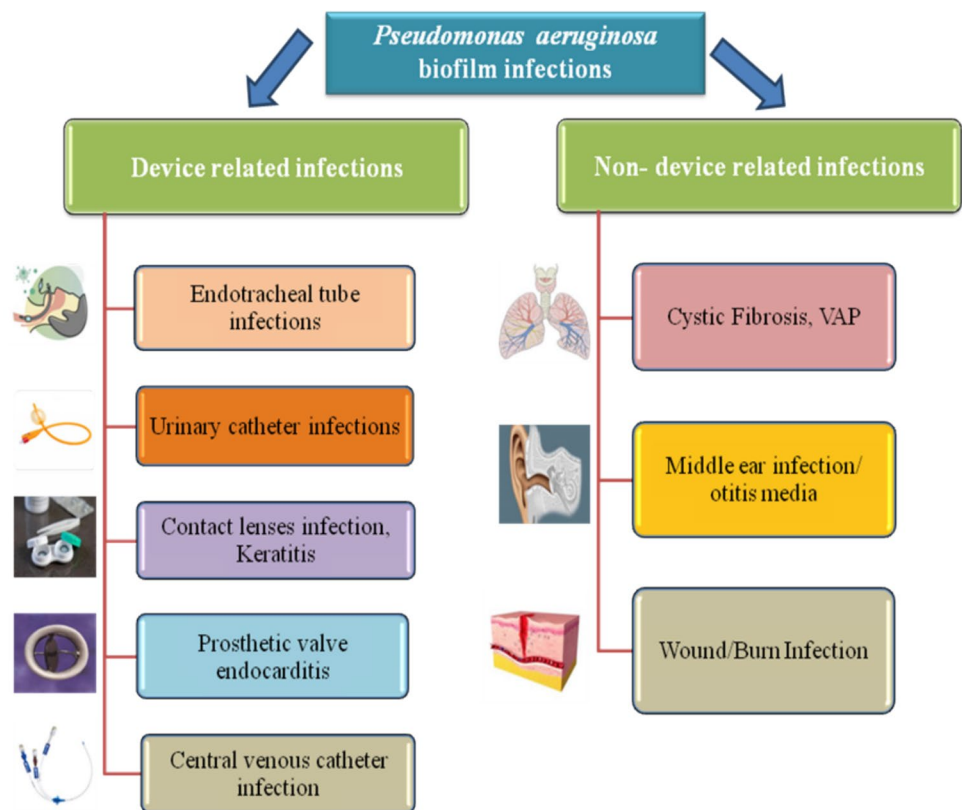
health concern, according to the International Nosocomial Infection Control Consortium (Rosenthal et al. 2016), having a prevalence of 7.1% to 7.3% (Magill et al. 2014).

Microbial biofilm can be effortlessly detected during the chronicity of diseases or infections, for instance- in patients with a chronic stage lung infection, wound infection, or rhinosinusitis (Chowdhary et al. 2016). This is due to a dense polysaccharide matrix, which leads to a persistent infection, the ineffectiveness of antibiotics, and escaping phagocytosis (Asati et al. 2017). This biofilm-producing pathogen is found to be mainly associated with infection related to healthcare (Fig. 1), such as infection of the urinary tract, cystic fibrosis, keratitis, VAP, ICU, surgical sites, otitis media, bloodstream infection, burn or wound (Ito et al. 2021; Kalluf et al. 2017; Ramos et al. 2013; Tuon et al. 2012).

According to Annette and Matthew et al. (2012), the most commonly used strains of *P. aeruginosa* are PAO1 and PA14. Initially, they were clinical isolates converted into laboratory strains to study the physiology and genetic aspects of *P. aeruginosa*. This pathogen requires a temperature range from 4 to 42 °C to survive. However, it can be overnight incubated at 37 °C. Luria Bertani (LB) broth is a favorable medium to culture *P. aeruginosa*; it also uses a wide range of compounds as sole carbon and nitrogen sources. Other media of choice to culture these bacteria specifically are *P. isolation agar* (PIA, Difco) and King A. PIA; Difco media selectively favors the growth of *P. aeruginosa*. At the same time, King A media enhances pyocyanin pigment production. The colonies of *P. aeruginosa* on Macconkey medium are lactose non-fermenting, giving positive results for the oxidase test. *P. aeruginosa* on nutrient agar displays a greenish color due to the production of pyoverdine pigment. A distinctly fruity smell is observed, and colonies are big, opaque, and flat with irregular borders.

Based on earlier research, Tang et al. (2017) employed an enhanced selective medium containing ceftriaxone, such as a Lemco-based selective medium (CTA1). This media encourages a better synthesis of pyocyanin pigment. Furthermore, *P. aeruginosa* produces more fluorescence at an incubation temperature of 42 °C when grown in selected media using King's medium B as the base (CTA2), which sets it apart from other *P. species*. Thus, *P. aeruginosa* can be diagnosed using the CTA2 medium. Unlike other bacteria, *P. aeruginosa* can break down acetamide and create ammonia as food. Based on this technique, Szita et al. (1990) created a liquid synthetic media that, compared to a culture medium enriched with ceftriaxone, allows for higher detection selectivity, sensitivity, and speed. Cultures of *P. isolates* are inoculated in several media containing biochemicals (like Triple Sugar Iron, peptone water, citrate, and urease).

Fig. 1 Biofilm-associated infections caused by *P. aeruginosa*. *P. aeruginosa* biofilm infections are categorized into device-related and non-device-related infections. The device-related infections include catheter-associated infections relating to the endotracheal tube, central venous catheter, peritoneal dialysis catheter, urinary catheter, vascular grafts, and prosthetic valve endocarditis as well as contact lenses infection. On the other hand, non-device-related infections include cystic fibrosis, middle ear infections, wound or burn infections, and keratitis. (Adapted from Tuon et al. 2022)



Biofilm production by *P. aeruginosa*

The lodgment of microbial cells in an extracellular matrix containing proteins, extracellular DNA, and exopolysaccharides, which gives promising protection to bacterial life, is known as a biofilm (Baker et al. 2001). Exopolysaccharides (EPSs), the principal constituent of microbial biofilm (Sutherland et al. 2001) aid in developing *P. aeruginosa* biofilm. Ryder et al. (2007) also observed that alginate, Pel, and Psl contribute to the EPS of *P. aeruginosa*. Among these, alginate is predominantly found within the biofilm of *P. aeruginosa* mucoid strains, providing biofilm maturation and stability. It attaches to the mucin in the respiratory tract and works as an adhesin protein for the attachment. This property of alginate is due to its highly viscous nature. The presence of acetyl groups favors enhanced viscosity, leading to the aggregation of water and nutrients.

Alginate protects microorganisms in biofilm by eliminating free radical oxygen secreted by the activated macrophages. Moreover, it shields the biofilm from phagocytosis. In this way, it allows the survival of bacteria and eventually leads to the pathogenesis of pneumonia infection. Alginate boosts resistance to antibiotics, as in the case of failed clinical treatments, it binds to aminoglycoside antibiotics, inhibiting their penetration into biofilm (Tuon et al. 2022). *Pel* gene influences the development of the biofilm

matrix's structure as it is rich in carbohydrates (Friedman and Kolter 2004).

Furthermore, alginate also serves a secondary function by boosting resistance to aminoglycoside antibiotics, a defense exclusive to the biofilm population (Heydari and Eftekhari 2015). Psl is a polymer that contains mannose and has an essential function in the initial stages of biofilm formation (Ma et al. 2006). In biofilm formation and stability, type II and III-secreted extracellular virulence enzymes are also involved (Moradali et al. 2017).

Biofilm formation

Biofilm is created when the planktonic bacteria first attach themselves to any biotic or abiotic surface by physical forces. This is followed by interaction between the planktonic microorganism and the medium to which it is fastened, which gets attached perpetually and encloses itself in an extracellular polymeric matrix, leading to the aggregation of cells. Biofilm formation is an irreversible process (Fig. 2). Later on, the development of microcolonies within a biofilm contributes to its maturation and growth by creating a 3-dimensional structure of matured biofilm. In order to spread the infection, these microcolonies are released into the environment to invade new adhesion sites (Jamal et al. 2018).

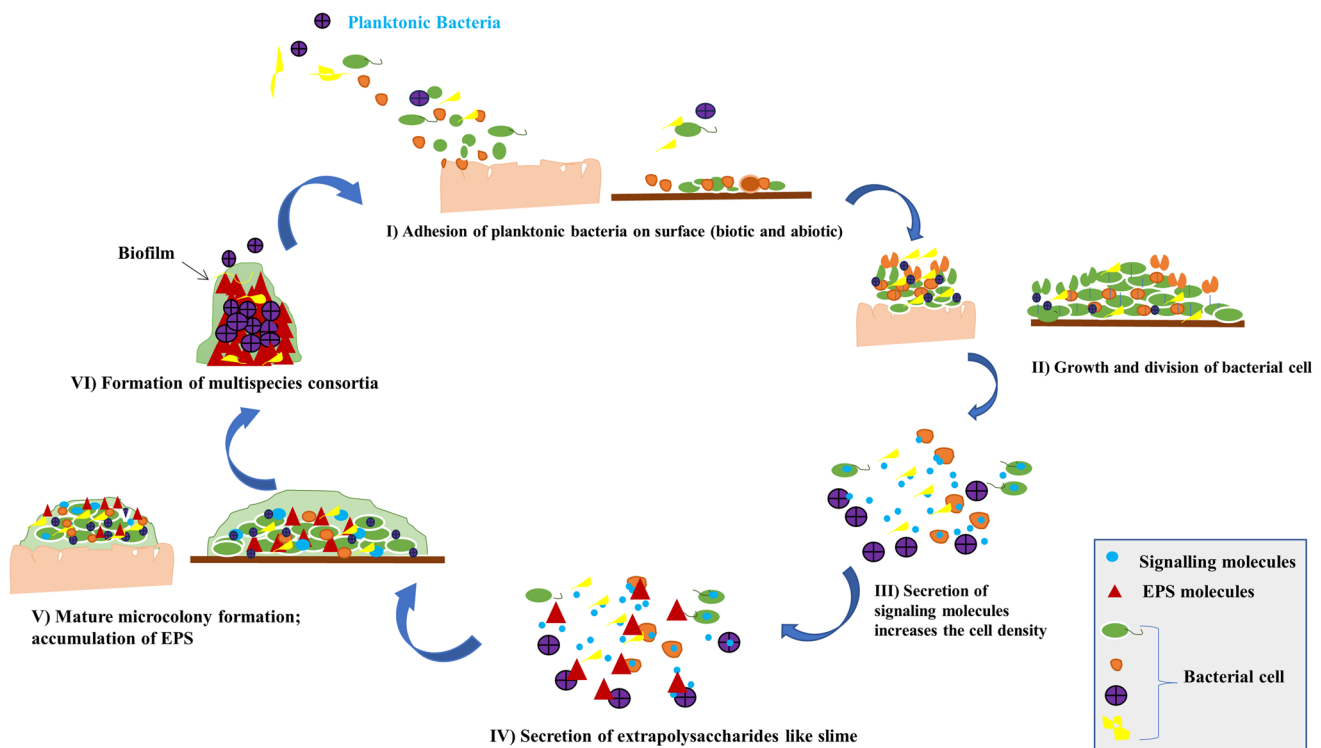


Fig. 2 Stages involved in biofilm formation. **I** Irreversible adherence of planktonic bacteria to abiotic/biotic surface. **II** Growth and division of bacterial cells. **III** Release of signaling molecules as a source of communication among the bacterial cells (referred to as quorum sensing) increasing the cell density gathering the cells in close proximities to each other. **IV** Secretion of extra polysaccharide, extracellu-

lar DNA by the microbial population forming the slime or matrix. **V** Multiple species of bacteria are attracted to the matured biofilm forming multispecies consortia. **VI** Formation of multispecies consortia; release of planktonic bacteria and repetition of the cycle on other places (adapted from Vasudevan (2014) J Microbiol Exp 1(3): 00014. DOI: <https://doi.org/10.15406/jmen.2014.01.00014>)

P. aeruginosa in a polymicrobial biofilm

By establishing collaborative or antagonistic interactions, the microorganisms efficiently react to one another, resulting in either suppression or stimulating effects. Two microorganisms interacting with one another (known as “interspecies or intraspecies interactions”) could affect how they live, and this will have a direct impact on how the infection spreads (Bisht et al. 2020). *P. aeruginosa* may also thrive in the biofilm alongside one or more bacteria, forming polymicrobial biofilm (PMB) or it can exist in solitary as such as monomicrobial biofilm (MMB). PMB of *P. aeruginosa* was observed by Bandara et al. (2020) and Price et al. (2020) with *Candida albicans* and *Staphylococcus aureus* respectively. Another PMB detected by Jordana-Lluch et al. (2020) consisting *P. aeruginosa*, *S. aureus*, *S. epidermidis*, and *Microrococcus luteus*.

The co-existence of *P. aeruginosa* and *S. aureus* is often chronic and persistent in patients with cystic fibrosis (CF),

diabetic foot ulcers, and other infections, leading to persistent colonization by microbial species and tardy wound healing synergistically (Alves et al. 2018). In furtherance, these organisms strengthen the bacterial invasion and impair the proinflammatory response in damaged keratinocytes (Fischer et al. 2021; Suryaaletha et al. 2018). *S. maltophilia* accelerates the expression of PQS genes, adhesion, and biofilm formation pathogenicity when found in PMB with *P. aeruginosa* (Rickard et al. 2006; Wang et al. 2016).

P. aeruginosa has also been commonly found with fungi forming polymicrobial biofilm. In a study by Briard et al. (2015) *Aspergillus fumigatus* growth has been shown to diminish when co-cultured with *P. aeruginosa*. It is also commonly co-isolated from CF patients along with *C. albicans* causing an opportunistic fungal infection that is present in burn wounds and the lungs of CF patients (Grainha et al. 2020; Hattab et al. 2022). PMB constituting *C. albicans*, *C. dubliniensis*, and *P. aeruginosa* increases the meropenem resistance in biofilm (Alam et al. 2020).

Biofilm regulation by quorum sensing

Like all living cells, prokaryotic cells also have a tendency to communicate with each other via signaling molecules. When the density of planktonic cells reaches a threshold as observed in biofilm formation, the signaling molecules released from the cells accumulate within the microbial community. These extracellular signaling molecules also known as autoinducers (AI) therefore alter the gene expressions including the genes that produce AIs (Kostakioti et al. 2013; Alav et al. 2018). This phenomenon is known as quorum sensing (QS) (Singh et al. 2021). Numerous processes are regulated by quorum sensing such as biofilm formation, sporulation, bioluminescence, antibiotic production, etc. (Rutherford and Bassler 2012; Williams and Camara 2009). Other than cell densities quorum sensing can also be triggered by alterations in pH, surrounding organisms, and rate of flow of signals (Soto et al. 2013; Geske et al. 2005).

The two primary quorum sensing systems in *P. aeruginosa* are Las and Rhl which detect the AIs and purely depend on them for their actions. Each system includes enzyme autoinducer synthases *lasI* and *rhlI* and corresponding transcriptional regulators *lasR* and *rhlR*, respectively. Additionally, the *P.* quinolone signal (PQS), a third QS system, and the integrated QS (IQS) system, a recently discovered fourth QS system, are present as QS systems (Sarabhai et al. 2016). They are linked together and exert intricate control over one another. These pathways aim to activate autoinducer (AI) molecules homoserine lactones (3O-C12-HSL and C4-HSL), PQS, and IQS, within the cells.

Among these quorum-sensing systems, the Las system is at the top of the cycle because of its hierarchical organization (Crespo et al. 2018) and detects 3OC12-HSL (N-3-oxododecanoyl-L-homoserine lactone), contributes as the foundation of the Las system. The Rhl system detects the signaling molecule N-butanoyl-homoserine lactone (C4 HSL). *LasR* is responsible for the activation of *rhlI* gene that makes Rhl system dependent on Las system (Henke et al. 2004). The PQS system, on the other hand, is dependent on signaling molecules from the class of 2-alkyl-4-quinolones (AQs) (Williams and Camara 2009). The QS system is responsible for the composition of the biofilm matrix (Allesen-Holm

et al. 2006; Sakuragi et al. 2007). QS regulates the synthesis of rhamnolipids, an extracellular surfactant produced by *P. aeruginosa*. Its synthesis creates atypical biofilm and exhibits improved surface dispersal (Lequette et al. 2005).

Inhibition of AHL signals suppresses the quorum sensing that shows an inhibitory effect on the biofilm. This is referred to as quorum quenching and the molecules showing such property of inhibition towards the signals are known as quenchers (Gilles and Tom 2015). As observed, the ratio of AHI molecules from both QS circuits (Las and Rhl) is the same in both biofilm and sputum from cystic fibrosis sputum (Singh et al. 2000; Sriramulu et al. 2005). Cystic fibrosis condition provides a similar atmosphere as biofilm rather than the planktonic counterparts. Therefore, this can be implemented to develop novel treatment strategies in cystic fibrosis patients.

Pathogenesis of *P. aeruginosa* biofilm

Microbial biofilm has a series of innate and acquired immune factors that aid pathogens inside biofilm to escape the host defense system and build an infection. Healthcare-associated transmission occurs mainly through direct and indirect contact. Improper hand hygiene is the cause of the spreading among healthcare workers and patients indirectly. Direct transmission occurs in contact with contaminated reservoirs or ingesting contaminated foods and water. The oral or respiratory route is the most common entrance for *P. aeruginosa*, and colonization occurs straightway after infection.

P. aeruginosa cannot alone manage to colonize the host. However, various virulence factors such as adhesions, the type III secretion system, and other secreted proteins help colonize and invade the host's immune system to cause infection (Streeter K et al. 2016) (Table 1). Once host invasion occurs, the development of *P. aeruginosa* infection most commonly progresses in three stages: (1) adhesion and colonization of bacteria, (2) local invasion, and (3) infectious diseases and systemic diseases.

The initial step before colonization and disease initiation is the adhesion of bacteria to the host cell (Pizzaro-Cerda et al. 2006). The single flagellum of the pathogen acts as the main adhesin, helping in motility, attachment, and biofilm

Table 1 Various virulence factors secreted by *P. aeruginosa* involved in biofilm production

Virulence factors	Functions
LPS (lipopolysaccharides)	Protect cell membrane; host cell poisoning; endotoxicity of lipid A; adhesion to host cell receptor; resistance against antibiotic; biofilm formation
OMP (outer membrane proteins)	Attachment, exchange of nutrients, resistance against antibiotics
Flagella, pili, adhesion	Antibiotic resistance; colonization, adhesion, swimming, and swarming in response to chemotactic signals
Exopolysaccharides, i.e., alginate, Psl, Pel	Biofilm formation and impaired bacterial clearance

development. The type IV pili facilitate the movement of bacteria while helping in biofilm formation and respiratory epithelial cell attachment (Bucior et al. 2012). Burrows et al. (2012) mention that the first step in biofilm development is binding mediated by pili. Motility is an essential factor aiding the virulence of pathogens and favoring the biofilm. He et al. (2022) in his work concluded that twitching motility and biofilm formation are correlated with each other.

This process often occurs when anatomical barriers are disrupted (skin, mucous membranes, etc.), as in the case of medical devices like central venous catheters (central venous, urinary, endotracheal). This pathogen possesses intrinsic resistance to various antimicrobial agents because of several factors. These factors are outer membrane permeability reduction, efflux pump expressing multidrug resistance, and enzymes inactivating antibiotic development (Hall et al. 2018; Cox et al. 2013).

Biofilm associated with indwelling medical devices (IMDs)

While hospital critical care units employ a variety of medical devices for patient care intervention and therapy, indwelling medical devices are commonly employed in all health setups. In order to administer fluids, blood products, drugs, and nutritional solutions and to monitor hemodynamics, catheters may be implanted. Several bloodstream and urinary tract infections that are associated with implanted medical devices are related to biofilm development (Patel et al. 2016; Jamal et al. 2018; Basher et al. 2021). Infections associated with IMDs, as seen in UTIs like urethral stents and sphincters, are riskier, as elucidated by Foxman et al. (2003).

Extensive use of medical devices in biomedical sciences makes patients more vulnerable to disease due to the colonization of devices with microbial biofilm causing device-related infection (DRI) (Singhai et al. 2012). Intravascular catheters, endotracheal tubes, indwelling urinary catheters, prosthetic joints, and contact lenses are some examples of indwelling medical devices (Garner et al. 1996). The prolonged usage of these devices for at least 48 h forms plasma protein coating on these devices and makes them favorable sites to harbor the pathogens initiating microbial colonization and biofilm formation as suggested by Arciola et al. (2018) and Zheng et al. (2018). Hence, increasing the incidence of DRI such as blood stream infections, catheter-related blood stream infections (CRBSI), catheter-associated urinary tract infections (CAUTI), and ventilator-associated pneumonia (VAP) (Horan et al. 2008). IUD (Intrauterine devices) users experience recurring infections and antibiotic resistance may be because of biofilm formation (Pruthi et al. 2003).

Staphylococcus aureus and coagulase-negative *Staphylococci* (CoNS) are among the most prevalent gram-positive

microorganism whereas gram-negative bacteria such as *Escherichia coli*, *Klebsiella pneumoniae*, *Proteus mirabilis*, and *P. aeruginosa* are hugely associated with biofilm in DRIs. Specifically, device-related biofilm infections occur in patients with loose immunity as well as prolonged hospital stays (Raad et al. 1998). *P. aeruginosa* isolated from catheterized patients produces a strong biofilm (He et al. 2022). One of the leading causes of ventilator-associated pneumonia (VAP) is biofilm, which grows very easily and quickly (within 24 h) on endotracheal tubes (ETTs) (Donlan et al. 2001). According to Sunarintyas (2018), the occurrence of *P. aeruginosa* in endotracheal tube biofilm has been highest followed by *S. aureus* leading to VAP.

Darouiche et al. (2004) noticed the prevalence of *S. aureus* and coagulase-negative *staphylococci* in two-thirds of medical device-related infections. The *S. aureus* forms biofilm clumps in the synovial fluid of the musculoskeletal joint and is easily observed through the naked eye. This occurs after the remodeling or replacement of these joints and provides the best example of biofilm production in prosthetic joints (Davies 2003). Furthermore, *Staphylococcus* species biofilm can also be seen infecting the mechanical heart valves such as prosthetic valve endocarditis and central venous catheters (Joo et al. 2012; Jamal et al. 2018; Veerachamy et al. 2014). Revidiwala et al. (2011) reported that Central Venous Catheters (CVCs) have a higher risk of device-related infection, with infection rates ranging from 3 to 5%.

Vickery et al. (2011) have observed the biofilm event in shunt placement during the post-operative phase is initiated by aggregation maximum number of skin commensals such as *Propionibacterium* and *Staphylococcus* species. Approximately 70% of infections occur within the 2 months following surgery, which strongly implies that colonization during shunt placement is a significant initiating event. Silicon or latex urinary catheters are extensively used by patients and harbor numerous harmful bacteria responsible for biofilm development. The most often isolated organisms from urinary catheters are *P. aeruginosa*, *S. epidermidis*, *S. aureus*, *E. faecalis*, and *E. coli* (Stickler et al. 2008; Luzum et al. 2020). *P. aeruginosa* along with *S. aureus* has been reported in contact lenses by Osei et al. (2022). Moreover, he updated in his study that the adsorption of a component of tear, glycoprotein Gp340 on frequently used contact lenses initiates the adhesion of these bacterial species thereby promoting biofilm formation.

Association of antibiotic resistance with biofilm formation

Antimicrobial substances function as stressors only when their value is below minimum inhibitory concentration (MIC). This may lead to more biofilm development, as Thi

et al. (2020) suggested. This may be possible, as Franci et al. (2018) stated because the drugs cannot adequately penetrate the biofilm matrix due to less diffusion. Bacteria may also change its physiological characteristics when present within biofilm communities. Some environmental factors may also contribute to this process. The antibacterial drugs are minimal, and the increasing rate of multidrug-resistant hinders their efficacy. Mekni et al. (2012) concluded that biofilms are protected from host immunity and antibiotic agents' effects, making them difficult to remove and posing a significant threat. Moreover, the biofilm matrix blocks the penetrating antibiotics inside the biofilm, thus preventing them from reaching the cells (Ansari et al. 2014). Clinical isolates' susceptibility to antibiotics is assessed using the disc diffusion method. A sterile loop is used to emulsify three to five well-isolated colonies from the organism's pure culture in three milliliters of sterile saline (0.9%). The turbidity of the suspension satisfies the 0.5 McFarland requirements. Antibiotic selection and antimicrobial susceptibility result interpretation adhere to the clinical and laboratory standard institute guidelines (CLSI, 2017).

In a study by M et al. (2021), *P. aeruginosa* isolates are MDR and biofilm producers with the highest resistance observed against piperacillin. Meanwhile, susceptibility was detected in fewer (24%) biofilm producers. Therefore, biofilm-producing isolates exhibit higher resistance compared to non-biofilm-producing isolates. Elmanama et al. (2013) suggested that the highest level of resistance against all antibiotics was used to determine the correlation between MDR and the biofilm-forming ability of microorganisms. Biofilm from urinary tract infections of non-catheterized and catheterized patients demonstrated by He et al. (2022) results in resistance to cefotaxime (100%) followed by ceftazidime, cefuroxime, and gentamicin. Analysis showed higher biofilm production in non-catheterized isolates than in catheterized isolates in multidrug-resistant (MDR) strains of *P. aeruginosa*.

Abdulhaq et al. (2020) reported a higher resistance rate against ceftriaxone in clinical isolates of *P. aeruginosa*, followed by meropenem, imipenem, and aztreonam. All MDR isolates showed biofilm formation in MPA (microtiter plate assay). Shrestha et al. (2019) also investigated the relationship between biofilm formation and MDR. The Kirby-Bauer disk diffusion method is used to detect the antibiotic resistance pattern. Susceptibility to polymyxin B and colistin has been determined. Hence, it can be assumed that multidrug resistance promotes biofilm formation in intensive care unit patients and patients using inpatient medical devices. Moreover, the predominance of isolates indicates resistance to commonly used anti-pseudomonal drugs and may pose a therapeutic challenge.

Telma et al. (2023) explained that clinical isolates of *P. aeruginosa* were predominantly resistant to ciprofloxacin. In

addition, a high susceptibility was marked against amikacin and tobramycin, and most isolates could produce biofilm. Karami et al. (2020) explored the susceptibility pattern in the biofilm produced by *P. aeruginosa*. Isolates were obtained from the environment as well as clinical settings. Antibiotic susceptibility testing through the Kirby-Bauer method determined the highest resistance against Ticarcillin/Clavulanic Acid. The maximum isolates were MDR, and half displayed strong biofilm production. Hence, MDR and biofilm production ability have also been associated in this case. *P. aeruginosa* frequently colonizes Foley's catheter tips along with CVCs verified by Veronica Folliero et al. (2021), who studied biofilm from indwelling medical devices. Polymicrobial growth has also been noticed mainly in CVC. The strains of this bacterium show significant resistance toward Fosfomycin and less towards piperacillin/ tazobactam (Veronica Folliero et al. 2021).

Gene regulation in biofilm formation

As discussed earlier when bacteria reach a critical concentration (i.e. quorum), they may receive information from other cells in the population and communicate with them to regulate the production of varying virulence factors through tiny signaling molecules known as auto-inducers. Sensing the critical concentration or quorum is known as quorum sensing (Rutherford et al. 2012). Quorum sensing controls multiple virulence factors and biofilm formation in *P. aeruginosa*, and these factors also promote the establishment and persistence of infection. Bacteria emit tiny chemicals known as acylated homoserine lactones (AHL), which travel to neighboring cells and trigger transcriptional regulators there. Three quorum sensing systems express toxic genes regulating virulence and biofilm production. Although coordinated gene regulation within cell populations is possible, the capacity of bacteria to generate signals is likely to give them a considerable advantage over their hosts. Recent studies have shown that biofilm formation by *P. aeruginosa* is regulated by the quorum sensing mechanism (QS). This process allows cells to communicate with one another to build biofilm and a variety of virulence factors (Ghanem et al. 2021).

Quorum sensing affects *P. aeruginosa*'s capacity to grow stalks, which resemble mushrooms. As a result, quorum-sensing mutants produce a thin, homogeneous coating on surfaces and regulate the synthesis of extracellular rhamnolipid surfactants, produced by *P. aeruginosa* (Davies et al. 1998). Quorum sensing influences biofilm formation and virulence factor production in burn wounds. As with *P. aeruginosa*, isolates carrying the *lasR* and *rhlR* genes appear to be important regulators of virulence factors and biofilm development, while the *lasR* gene positively regulates biofilm development, proteolytic action,

pyocyanin production, and rhamnolipid biosynthesis. Protease and rhamnolipid synthesis are positively impacted by the QS regulating *RhlR* gene (Elnegery et al. 2021). Specific genes, such as *algD*, *pslD*, *pelF*, *Ppgl*, and *PAPI-1*, also contribute to biofilm production (Rajabi et al. 2022). Genes such as *pslA*, *lasR*, and *pelA* have been majorly involved with biofilm-forming strains of *P. aeruginosa* (M et al. 2022; Abdulhaq et al. 2020).

PpgL protein or periplasmic protein functions as gluconolactonase in the periplasmic area of bacteria. The *PA4204* gene, which codes for this protein and has gluconolactonase activity, is essential for healthy bacterial growth and biofilm development in the periplasm. According to predictions, this protein is a member of the propeller fold- β family, which has several cellular activities and a usually compact shape (Song et al. 2019). The *PAPI-1* gene is implicated in numerous cellular infections and has a significant role in etiology and disease dissemination. As a result, people with cystic fibrosis may be more susceptible to chronic infections (Carter et al. 2010; Harrison et al. 2010; Saravanamuthu et al. 2004). *PAPI-1* encodes *PvrR*-like regulatory genes, which implicate in biofilm development and resistance. The synthesis of several virulence factors, such as elastase (*lasB*), *toxA*, and *lasA* protease, is regulated by *C12-HSL/lasR* (Bleves et al. 2005). Pyocyanin synthesized by *P. aeruginosa* indicates *rhl* and *PQS* system activity, as observed in numerous investigations (Lee et al. 2015; Reuter et al. 2016).

Furthermore, Doltshah et al. (2021) found that all isolates carried the *cupA* gene and seven quorum sensing genes such as *rhlI*, *rhlR*, *rhlAB*, *lasB*, *lasR*, *lasI*, *aprA*, *pslA* gene. A strong correlation exists between the *rhlA* gene and integron (class 1 and 2) and biofilm formation. PCR was used to identify integrons, the *lasB* gene, and *rhlA* in biofilm development (Karami et al. 2020).

Prevention of biofilm and anti-biofilm agents

Biofilm infections linked to medical devices are quite concerning. One excellent way to stop these diseases is to produce materials resistant to biofilm colonization. Some anti-biofilm agents are mentioned in Table 2. Using materials embedded with nanoparticles, modifying the nanostructure of materials, or impregnating devices with antimicrobial chemicals are some methods to reduce the susceptibility of materials to biofilm attachment (Masak et al. 2014). The early phases of biofilm formation are represented by the creation of conditioning films and the initial attachment of bacteria to catheter surfaces; effectively controlling these processes should prevent biofilm formation associated with catheters (Pelling et al. 2019). Changing a catheter's hydrophobicity, charge, or topology or adding antimicrobial medicines meant to destroy colonizing microorganisms are two ways to prevent these initial stages of catheter colonization (Francolini and Donelli 2010).

Antibiotics are efficient only if the pathogen has not progressed into its resistant form. Antibacterial drugs like Tobramycin work quite efficiently against *P. aeruginosa* biofilm infection (Yang et al. 2023). Anti-virulence medications target quorum sensing as well. These medications seek to obstruct signal/receptor connections or degrade quorum-sensing signals and prevent *P. aeruginosa* cells from communicating with one another (Diggle et al. 2007). Azithromycin targets virulence factors and shows better clinical results in cystic fibrosis patients (Saiman et al. 2003). Another method of QS suppression is treatment with quorum-quenching enzymes, such as lactonases and acylases. These oxidize AHLs, also known as oxidoreductase, which alters AHLs (Bzdrenga et al. 2017). Some enzymes have beneficial effects, such as anti-biofilm on wound grafting. These include collagenase, trypsin, fibrinolysin, lysozyme,

Table 2 Different methods for prevention of biofilm and their mechanism of action

Category	Mechanism of action	Example	Reference
Antimicrobial agents	Antibacterial activity will be enhanced in an acidic medium	Tobramycin	Yang et al. (2023)
Antimicrobial peptides	Bind to the cell surface; penetrate into the bacterial cell; depolarize; cell death	PLG0206 (WLBU2)	Di et al. (2020)
Bacteriophage	Induce lysis of bacterial cells, reduce the biofilm in implants; lysis of cell	Bacteriophage delivering hydrogels	Wroe et al (2020)
Enzyme	Degrade the polysaccharide in EPS of biofilm	Mixture of carbohydrase enzymes from <i>Aspergillus niger</i>	Kaur et al. (2021)
Essential oil	Destroy 80% of biofilm; decrease EPS, swarming, and rhamnolipid	Murraya koenigii	Sankar et al. (2015)
Immunotherapy	Affect the biofilm genes <i>algD</i> and <i>Pipier</i>	<i>E. coli</i> Nissle1917 outer membrane vesicles	Alei et al. (2023)
Nanoparticles	Oxidative stress, cell lysis, colonization prevention oxidative stress, inhibition, genetic mutation, structural alteration	Agnosy (Silver nanoparticles) Stimuli-responsive NPs: SPIONs	Ansari et al. (2015) Belkahla et al. (2019)

streptokinase, bromelain-derived debris, and dispersion B. These enzymes can also eliminate obstacles that hinder wound healing, such as weaker tissues, bacterial biofilm, and scars (Vetrivel et al. 2021). A cocktail of enzymes found in the *Aspergillus* genus has been found to cause the degradation of extrapolysaccharide (Kaur et al. 2021).

As signaling within the biofilm formation and its spread resembles the c-di-GMP signaling pathway, a proposal of novel therapeutics against biofilm, such as suppressing the signal, can be employed. Nevertheless, doxorubicin (formerly known as a potent c-di-GMP inhibitor) has been used in a study that unexpectedly revealed an increase in biofilm growth (Groizeleau et al. 2016). Even with these developments, more research is required to determine whether inhibiting c-di-GMP signaling could be therapeutic. Multiple organisms produce peptides having antimicrobial qualities and synthetic antimicrobial peptides have been created, which are potent antibiofilm agents (Hancock et al. 2016). Certain peptides inhibit bacterial cell adhesion, impeding bacterial biofilm production (Overhage et al. 2008). PLG0206 (WLBU2) antipeptide attaches to the bacterial cell surface and causes depolarization of the cell (Di et al. 2020). In the race of antibiofilm agents, bacteriophages have proved their efficacy specific to implants as they cause cellular lysis in the biofilm (Wroe et al. 2020; Amankwah et al. 2022).

Recent technological developments such as nanotechnology have made it possible to use novel therapy strategies to address the issues (Ansari et al. 2015; Belkahla et al. 2019). An emerging technology called CRISPR-Cas (clustered regularly interspaced short palindromic repeats-CRISPR associated protein) such as *P. aeruginosa*'s type 1-F CRISPR-Cas system plays a crucial part in several alternative functions, including phage resistance, biofilm suppression, and impairment of swarming motility (Gholizadeh et al. 2017; Heussler et al. 2015). Phage therapy also aids in *P. aeruginosa* eradication on the decline of CRISPR-Cas defense systems by QS inhibitors (Gholizadeh et al. 2017).

Methods of biofilm detection

For the purpose of detection the biofilm, various phenotypic and genotypic techniques have been employed so far. Biofilm can be examined using electron microscopy and molecular techniques such as polymerase chain reactions that are more focused and sensitive. However, these techniques are cost-effective and need specialized human resources (Achiras et al. 2020). The phenotypic methods are quick, easy, and relatively sensitive; identifying the biofilm-forming characteristics. These methods are categorized into qualitative-tube method (TM) modified Congo red agar (MCRA), and quantitative method-tissue culture plate method (TCPM) (Knobloch et al. 2002; Basnet et al. 2023). The tissue culture

plate (TCP) or microtiter plate assay (MTPA) approach is the most established and dependable screening technique to detect the biofilm than the tube method (TM) and congo red assay (CRA) (Harika et al. 2020; Maione et al. 2023). Several researches have favored the MTP assay as the best phenotypic method for biofilm detection (Borel et al. 2023; Knobloch et al. 2002; Valadbeigi et al. 2023; El-Hossary et al. 2023).

The bacteria-causing device-related infections have been associated with biofilm production which is an important breakthrough in the health sciences (Singhai et al. 2012). When advanced microscopy techniques are not available, biofilm growth can be frequently proven in clinical practice by surface adherence on a bioprosthetic surface using the tube approach. The observations made by Singhai et al. (2012), in their study, concluded that all device-related isolates produced biofilms in varying degrees and there was a good correlation between the scanning electron microscopy and tube technique biofilm detection. Using the tube approach, *S. aureus*, *P. aeruginosa*, and *E. coli* demonstrated a moderate capacity to produce biofilms. Valdivieso et al. (2023) in their work used confocal laser scanning fluorescence microscopy (CLSM) to visually inspect the bacterial morphology and the three-dimensional arrangement of cells within the dual biofilm before and after expansion, comparing them with non-expanded biofilms seen with scanning electron microscopy (SEM) in order to assess the effect of the expansion treatment on the integrity of biofilms. The existence of multi-species biofilms has also been revealed on urinary catheter surfaces by Donelli and Vuotto (2014) combining culture methods with field emission scanning electron microscopy (FESEM) observations.

Microtiter plate assay (MTP)

In order to investigate the bacterial attachment, the microtiter plate method was developed by Madilyn Fletcher (Fletcher et al. 1977). The capacity of the cells to attach to the PVC microtiter plate (MTP) wells served as a biofilm development monitor (Labware by Beckon Dickson) (Pruthi et al. 2003). This method is best for the quantification of biofilm depending on the absorbance of the isolates.

In this detection assay, the wells are emptied and washed to remove planktonic cells at different time intervals before staining the biomass attached to the surface of the wells. Biofilm biomass can alternatively be quantified by detachment and subsequent plating. In the microtiter plate assay, the biofilm biomass is assessed by measuring all attached biomass (Joana et al. 2017). This procedure uses 96-well microtiter plates containing LB broth in which the production of biofilm is detected (Stepanović et al. 2007). 200 µl of the already identified bacterial strains are diluted in the ratio of 1:100 and inoculated in each well of the plate containing

LB broth (beginning turbidity of 0.05 at 600 nm). The plates are well placed in an aerobic incubator (Fisher Scientific, USA) and incubated aerobically for 24 h at 37 °C. The wells' contents are decanted after incubation. Then 0.1% crystal violets are applied to the adherent biofilm layer that had developed in each MTP well and left for 15 min at room temperature. Following staining, the excess stain is removed by rinsing the MTP in a large water container and aspirating the dye with a pipette. According to Stepanovic et al.'s (2007) criteria, strains are categorized into four groups for more straightforward result interpretation: none, weak, moderate, and vigorous biofilm producers. The intensity of isolates to produce biofilm is classified into four categories according to the criteria opted by Rehman et al. (2022). On the basis of optical densities (OD), the biofilm producers are identified as strong, moderate, weak, and non-biofilm producers (Kamali et al. 2020).

- i) Non-biofilm producer: if the optical density of the isolate is less than the cut-off value (OD_c), i.e., $OD < OD_c$.
- ii) Weak biofilm producer: if the optical density came out is more than OD_c but less than twice of OD_c, i.e., $OD_c < OD < 2 \times OD_c$.
- iii) Moderate Biofilm producer- If the optical density is more than twice OD_c but less than four-fold OD_c, i.e., $2 \times OD_c < OD < 4 \times OD_c$.
- iv) Strong biofilm producer: if the optical density is greater than four-fold OD_c, i.e., $4 \times OD_c < OD$.

Where, OD_c = average OD of negative control + (3 × Standard deviation of negative control) (Kamali et al. 2020).

Congo red agar

Freeman et al. (1989) initially described a method for qualitatively detecting biofilm production using Congo red agar (CRA) medium. Dry crystalline black colonies indicate biofilm production. Reid et al. (1999) employed this method, which involved the preparation of Congo red stain as a concentrated aqueous solution. After autoclaving this stain at 121 °C for 15 min, it was added to the brain heart infusion agar with sucrose at 55 °C. This test utilizes Congo red dye as a pH indicator, showing black coloration between 3.0 and 5.2 at pH. Congo red agar medium was inoculated with the culture and incubated in an aerobic environment for 24 to 48 h at 37 °C. After this period, colonies exhibiting dark red or blackish along with dry or crystalline consistency are considered as biofilm producers, whereas colonies expressing red color with smooth and darker appearance in the center are considered as non-biofilm producers (Lima et al. 2017). Solati et al. (2015) have also used this method to observe the biofilm production by *Staphylococcus epidermidis* in urine

and blood isolates of patients suffering from urinary tract infection and septicemia, respectively. It was investigated that 55% of the isolates had the highest potential for forming biofilm, with most of the isolates obtained from urine (Solati et al. 2015).

Crystal violet dye

Crystal violet (CV) dye is used in microplate testing to determine the total biomass of the biofilm, additionally to tube tests for biofilm detection. It is an inexpensive material, and the test may be readily interpreted and replicated several times (Shukla and Rao 2017). Nevertheless, the drawback of this dye is that it stains the entire structure of the biofilm uniformly. It is the most popular colorimetric test applied in microplate assays as it allows the measurement of the total biomass and biofilm matrix and easier to calculate the absorbance (Di Luca et al. 2017). Numerous research studies have been reported using crystal violet to identify and measure the development of microbial biofilm (Xu et al. 2016; Vuotto et al. 2017; Mohammadi et al. 2020).

Safranin dye

Safranin is a non-toxic dye utilized in a manner akin to crystal violet. It yields lower optical densities than crystal violet staining, so it is less commonly utilized in biomass estimation (Ommen et al. 2017). Safranin has been employed as a dye in the microplate test by several authors, like Babapour et al. (2016) and Karami et al. (2020), to identify clinical isolates of biofilm producers.

Light or fluorescent microscopic examination

The optical potential of light microscopy can be enhanced by including intrinsic or additional fluorescence emissions, a technique employed by fluorescence microscopy (Wilson et al. 2017). Fluorescent dyes contain fluorescent molecules known as fluorophores that are used in these microscopes in such a way that when these molecules are integrated into biological structures, they liberate light. This released light forms images to examine biofilm properties, such as the viable cells occupying space and their morphological and functional features throughout the growth or treatment stages (Heydorn et al. 2000). Moreover, cell fluorescence can also be achieved with the help of a genetically modified organism expressing a fluorescent protein. Incorporating foreign DNA into the bacteria genetically to produce luminous gene products is another method of causing cell fluorescence. A plasmid, or little foreign DNA, is most frequently used to accomplish this, as foreign DNA incorporated into the bacterial genome could be accommodating in tracking the communicating genes (Chalfie et al. 1994). Although these

approaches require more time to prepare a sample (biofilm staining) or the method (generating a fluorescent cell), additional knowledge aids in understanding the details of life inside biofilm (Chimileski et al. 2014; Heydorn et al. 2000).

Tube adherent method

In the tube adherent method, culture is formed in a polystyrene tube with a round bottom and stained with 0.2% crystal violet such that the biofilm formed is inspected for the intensity of staining and interpreted as weak, moderate, firm, or no biofilm producer (Kunwar et al. 2021). The tubes are incubated for 24 h at 37 °C. This assay uses plastic tubes holding sample inoculums dyed with 1% crystal violet. The tubes are inverted in an adsorbent paper after washing with phosphate buffer saline and fixing it with sodium acetate. They are observed for a stained film on the tube walls and bottom, which denotes biofilm formation (Halim et al. 2018). The thick layer visible on the tube wall and bottom is considered a strong biofilm, while the thin layer present on the bottom of the tube signifies moderate biofilm (Daouadji et al. 2021). The absence of any visible layer on the tube is considered harmful for biofilm formation. This experiment is performed in triplicates (Mishra et al. 2015).

Fluorescence in situ hybridization (FISH)

This technique involves attaching short oligonucleotides labeled with fluorescence to the ribosomal RNA of the intended host species. Its benefit is that samples can be analyzed using microscopy without needing previous treatment, making it simple to identify microbial aggregation (Frickmann et al. 2017). Bernardi et al. (2019) detected and quantified the biofilm with the help of confocal laser scanning microscopy (CLSM) and the fluorescence in situ hybridization (FISH) method. Biofilm-forming strains of *Fusobacterium nucleatum* and *Streptococcus* sp. were identified by them in the dorsum of the tongue (Bernardi et al. 2019). Additionally, compared to other widely used approaches, this approach has been tested in bacterial vaginosis and has been shown to identify biofilm-forming bacteria in vaginal samples with higher specificity (Machado et al. 2015).

Electron microscopy

According to Costa-Orlandi et al. (2017), the most popular techniques for seeing biofilm are transmission electron microscopy (TEM), scanning electron microscopy (SEM), and confocal laser scanning microscopy (CLSM). Because of the incredible resolution of this microscopy, all biofilm structures can be analyzed, making it easier to directly detect and visualize the biofilm and its origin, such as in a catheter (Sugimoto et al. 2016; Schlafer and Meyer 2017). With

the use of scanning electron microscopy (SEM), the surface topography is captured in high resolution. The total magnification of this method can be between 10 and 500,000 times, which makes it extremely useful for analyzing tiny structures, such as biofilm (Adetunji et al. 2012; Arnold et al. 2000). SEM allows for gathering high-resolution pictures that help assess bacterial interaction, EPS organization, and biofilm morphology. This helps improve understanding of how biofilm forms and persists (Baum et al. 2009; Hung et al. 2013). Samples can be dehydrated, fixed, and stained using SEM; however, during all of these procedures, particularly the dehydration step, the shape and structure of the biofilm may change. This is a drawback of the method (Costa-Orlandi et al. 2017). Transmission electron microscopy (TEM) is preferred over SEM as it visualizes biofilm architecture more prominently. The extracellular polymeric matrix of the biofilm is more stable when the microbial aggregation is coated with resin.

The fact that the biofilm topography cannot be visualized is a drawback of TEM (Mohammed et al. 2017). Confocal laser scanning microscopy (CLSM) is another imaging technique to create finer, detailed 3-D biofilm images. Surrounding the sample, the region is concentrated and produces well-resolved 2-dimensional pieces integrated into the final 3-dimensional representation. It can simultaneously examine numerous fluorescent markers in succession using one or more moving sensors. Integrated microscopes can initially range in price from hundreds to thousands of dollars, depending on the system setup. This technique ensures 3D visualization of the entire biofilm architecture. CLSM uses fluorescent help in marking the several contents of biofilm along with the extracellular matrix and spatial association of the biofilm. In addition, macromolecules such as proteins, nucleic acids, lipids, and polysaccharides can be easily identified (Costa-Orlandi et al. 2017). For accurate measurement and result inspection, this technique also needs workers with training and expertise (Wilson et al. 2017). The phenotypic characteristics and morphological differences of microbial biofilm are becoming increasingly apparent through CLSM.

Polymerase chain reaction (PCR)

Based on amplifying the gene sequence specific to the pathogen, this technique helps in detecting the pathogens. This method is extremely specific and sensitive in the identification of genes involved in biofilm formation (Rajapaksha et al. 2019). The three biofilm-encoding genes, *algD*, *pslD*, and *pelF*, are assessed in all *P. aeruginosa* isolates using the polymerase chain reaction (PCR) technique using particular primers (Kamali et al. 2020). Different genes associated with biofilm formation are mentioned in Table 3. Wedad et al. (2020) used a real-time PCR cycle; in which four negative control samples (one for each gene) were included with

Table 3 Association of various factors with biofilm production by *P. aeruginosa*

Biofilm detection	Antibiotic resistance/susceptibility	Gene regulation	Conclusion	Researcher
Congo red agar/microtitre plate assay	Ceftriaxone, meropenem, imipenem, aztreonam	<i>pslA</i> gene	MDR and <i>pslA</i> gene associated with Bio-film;	Abdulhaq et al. 2020
Microtitre plate assay	Cefotaxime, ceftazidime, cefuroxime, ciprofloxacin	<i>VIM</i> gene and <i>PSE</i> gene	MDR isolates association with catheterized patients	He et al. 2022
Microtitre plate assay	Ceftazidime	<i>lasR</i> , <i>rhlR</i> gene	<i>LasR</i> regulates biofilm formation	Elnegeery et al. 2021
Microtitre plate assay	Ticarcillin/clavulanic acid, aztreonam	<i>lasB</i> , <i>rhlA</i> gene detected	Positive correlation of genes with biofilm	Karami et al. 2020
Tissue culture plate method	Biofilm was resistant to ciprofloxacin, cefepime, colistin, norfloxacin, aztreonam, imipenem	<i>lasI/lasR</i> and <i>rhlI/R</i> detected	QS genes are associated with biofilm formation	Ghanem et al. 2023
Microtitre plate assay	Biofilm susceptible to ciprofloxacin, amikacin, vancomycin, tetracycline, gentamycin, erythromycin	<i>lasR</i> and <i>lasI</i> genes	MDR and QS genes regulated biofilm production	Mohammad AS et al. 2023
Microtitre plate assay	No susceptibility pattern observed	<i>lasI</i> , <i>lasR</i> , <i>rhlI</i> and <i>rhlR</i> genes	Only the genes associated with biofilm	Karuuna O et al. 2010
Microtiter plate-based assay	Susceptible to amikacin and tobramycin; resistant against piperacillin/tazobactam, ceftazidime and gentamicin	<i>toxA</i> and <i>lasB</i> genes	MDR and virulence genes associated with Biofilm formation	De Sousa et al. 2023
Microtitre plate assay	Susceptible to tobramycin, gentamycin	<i>ndvB</i> , <i>tssCI</i> , <i>PA1875-1877</i>	Three novel biofilm-associated genes detected	Zhang et al. 2013
Microtitre plate assay	Resistant to ceftoxime, ceftazidime, meropenem, imipenem, piperacillin	<i>bla_{CTX}</i> , <i>bla_{SHV}</i> , <i>algD</i>	ESBL genes along with strong biofilm-producing genes detected	Ghasemian et al. 2023
Microtitre plate assay	Resistant to gentamicin, amikacin, amoxicillin clavulanic, ertapenem, imipenim	<i>algD</i> , <i>pslD</i> , <i>pelF</i> genes	An elevated multidrug resistance association with biofilm formation	El Sayed et al. 2021
Microtitre plate assay	Resistant to gentamicin, imipenim, tobramycin, piperacillin	<i>PslA</i> gene	Biofilm associated to increase in resistance as well as <i>Psl</i> gene	Emami et al. 2015
Crystal violet assay	Resistant to ampicillin, nalidixic	<i>PslB</i> , <i>PslA</i> , <i>algD</i> , <i>pelD</i>	Synergistic effect of MDR with biofilm-producing genes	M. Divyashree et al. 2022
Confocal laser scanning microscope	Resistant to tobramycin	<i>asrA</i>	MDR is not associated with biofilm. Novel approach for biofilm detection	Musken et al. 2010
Confocal laser scanning microscope	Resistant to colistin, polymyxin B	<i>parR</i>	MDR was found associated with biofilm production	Dotsch et al. 2012

deionized water and used in place of the template. In a work performed by Doltshah et al. (2021) PCR was used to assess genes such as *cupA*, *pslA*, *rhlI*, *rhlR*, *rhlAB*, *lasB*, *lasI*, *lasR*, and *aprA* involved in biofilm production. Although these techniques require less time in presenting results, they are not effective in providing information regarding the actual biofilm flora present on medical device samples (Bryers et al. 2008).

Omics analysis of microbial biofilm

Molecular analysis primarily explores the specific primer and target gene associated with it which limits the zone of knowledge of the entire biofilm. On the other hand, newly developed technologies such as omics are paving the way to acknowledge the whole microbiome of biofilm by deeply digging into the transcription, metabolism, and translation processes within the microbial biofilm (Chiu and Miller 2019). This technology comprises transcriptomic, proteomic, and metabolomic assays. Analysis of transcription within omics includes express sequence tag, serial gene expression analysis, DNA microarrays, and RNA-Seq (Hrdlickova et al. 2017). Proteomics consists of various methods of detection such as 2D gels (DIGE), iTRAQ (gel-free), spiked-in peptides, and label-free quantitation (Otto et al. 2012). Metabolomics includes mass spectroscopies such as TOF-MS and capillary electrophoresis (Chiu and Miller 2019; Washio and Takahashi 2016; Caputo et al. 2018).

Matrix-assisted laser desorption ionization time of flight mass spectrometry or Maldi-TOF an omic tool for microbial biofilm detection compares the intensities of microbial biofilm into strong and weak on the basis of molecular masses. However, there are certain limitations as discussed in different literatures that makes the MALDI-TOF out of the way of being the gold standard (Singhal et al. 2015; Mlynarikova et al. 2016).

In-vivo models for biofilm detection

Studies of biofilm-related diseases have been feasible on in-vivo models. For instance, chinchillas are used to study otitis media biofilm, pigs for wound biofilm, etc. (Bjarnsholt et al. 2013). Selection of in-vivo models is based on their physiology and anatomical structure related to the specific disease and the metabolic activity and genetic composition are assessed through omics analysis (Rumbaugh et al. 2023). Pathogenesis of the disease as well as the immune response of the host against the biofilm development are better demonstrated in in-vivo models (Bjarnsholt et al. 2013). The efficacy of antimicrobial agents can also be determined on the in-vivo models (Rumbaugh et al. 2023). Meanwhile, there

are a few drawbacks such as slow rates of ethical approval and small biofilm architecture compared to the in vitro study (Silva et al. 2021).

Conclusion

This review deliberates the characteristics of *P. aeruginosa* associated with biofilm formation. The correlation of biofilm with antimicrobial resistance and quorum sensing genes majorly contributes to biofilm production. Also, by reviewing this pathogen's general characteristics, we can easily assume the cause for inhabiting biofilm communities is its high prevalence rate, nosocomial nature, and frequent encounters with indwelling medical devices. Biofilm enhances the resistance against several irradiation treatments, disinfectants, and the immune system *P.* infections may become potentially multidrug-resistant which may later become difficult to treat. Therefore, it is crucial to observe the various patterns of susceptibility as it can contribute to eliminating the impediment in the effective management of chronic infections caused by biofilm. Hence, it will be feasible to understand the effect of a diverse group of antibiotics and their resistant pattern on established microbial biofilm to avoid inappropriate treatment. Various methods of biofilm detection have been discussed in this review as there is a lot of gap in the knowledge of more appropriate tools for detecting a biofilm. Although the intensity of biofilm can be determined through microtiter plate assay, gene analysis through PCR and visual demonstration of biofilm through electron microscopy are the best diagnostic tools for biofilm detection because of the high specificity and sensitivity along with less time used. Due to limited research on in vivo models the range of biofilm detection is not optimum. An awareness of antibiotic resistance profiles allied with biofilm detection will aid in providing more appropriate treatment that will eventually improve patient management, invigorating the scientific community to search for novel treatment strategies to combat the threats. These upgraded novel strategies should be explored greatly to combat biofilm infections.

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